

Discharging H-diag IV: the full off-diagonal operator norm – formulation,

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope (honest verdict)

The constant certificate (hdiag-offdiag-constant-certificate) bounded the CONDENSATE-DIRECTION ratio $R_{\text{lead}} \leq 0.174$ at the operating intensity and named the FULL worst-direction operator norm as the residual. The operator directed the next step: construct the complete off-diagonal Hessian and bound $\|E^{-1/2}B_{\text{od}}E^{-1/2}\|_{\text{op}} < 1$ for the lattice class. This note executes that step and reports the honest verdict: **the full norm does not close at second-cumulant order; it merges with RES-5**. The document is therefore a FORMULATION (the Hessian structure + the route + the merge point), not a certificate – naming it a “certificate” would overclaim.

2. The full off-diagonal Hessian

$\text{Hess } F[G_*] = E + B_{\text{od}}$, with the entropy curvature $E = \frac{1}{2}G_*^{-1} \otimes G_*^{-1} \succ 0$ (positive-definite) and the interaction Hessian $B_{\text{od}} = \text{Hess } \Phi|_{G_*}$. In the off-diagonal (Bloch, reciprocal-transfer Q) sector B_{od} has three parts:

- **Hartree** – depends on G only through $M_{\text{tot}} = \text{Tr } G$, hence NO off-diagonal block (convex, $\Phi''_{\text{diag}} = \frac{3}{2}u_{\text{eff}} > 0$, companion note);
- **Exchange (Fock)** – $\propto u_{\text{eff}}$ times the off-diagonal $\int \int G(x, y)^2$ kernel: this is the load-bearing off-diagonal block;
- **Condensate** – $\propto u_{\text{eff}} A^2 p_2(Q)$ (the Math428 $W(Q)$ source), the additive-energy-weighted coupling.

The worst-direction certificate $\|E^{-1/2}B_{\text{od}}E^{-1/2}\| < 1$ is equivalent to $E + B_{\text{od}} \succeq 0$ over ALL off-diagonal δG , i.e. the entropy curvature dominating the (possibly negative) exchange block.

3. Why $u_{\text{eff}} > 0$ does not suffice (the obstruction)

The Hartree convexity established that F_0 (condensate-free) is convex because $\Phi''_{\text{diag}} = \frac{3}{2}u_{\text{eff}} > 0$. But the EXCHANGE block carries the opposite sign of the direct term: for a repulsive interaction the Fock exchange is ATTRACTIVE. Here the bare quartic is itself attractive ($u = -0.86 < 0$) and only the sextic dressing lifts the EFFECTIVE direct coupling to $u_{\text{eff}} = +2.685 > 0$; the exchange block’s sign is therefore NOT fixed by $u_{\text{eff}} > 0$. Consequently $B_{\text{od}} \succeq 0$ is NOT guaranteed, and the second-order envelope R_{lead} (which uses the bare entropy propagator E^{-1} , i.e. assumes $E + B_{\text{od}} \approx E$) does NOT bound the full norm. The full norm requires the exchange-Hessian eigenvalues.

4. The Gershgorin route and what is controlled

A diagonal-dominance (Gershgorin) certificate would give $E + B_{\text{od}} \geq 0$ if, per channel Q , the entropy curvature dominates the row-sum $\sum_{Q'} |B_{\text{od}}(Q, Q')|$. The row-sums ARE additive-energy objects: the channel overlaps are bounded by the pair multiplicities p_2 , so $\sum_{Q'} |B_{\text{od}}(Q, Q')| \lesssim$ (the same $E_+ = \sum_Q p_2^2$ controlled by R-025/R-026; verified: BCC max row-sum $48 \leq NE_+$). Thus the COMBINATORIAL part of the Gershgorin bound inherits the lattice control already used for STEP-5B (Gershgorin reduction) and H-ADM-COH (additive energy). **The missing input is the analytic SIGN/magnitude of the exchange block per channel** – not a counting quantity, but a loop (bubble) integral with a sign, which at the dressed level is a beyond-second-cumulant object.

5. The merge with RES-5

The exchange-Hessian eigenvalues, and the self-consistent $(E + B_{\text{od}})^{-1}$ resummation, are exactly the corrections that the matched-second-cumulant bookkeeping (SC-SCOPE) holds fixed. They live at the same order as RES-5 (matched-order to exact / GAP-2). Therefore the FULL off-diagonal operator-norm certificate MERGES with RES-5: it cannot be closed within the second-cumulant scope, and its closure is the same analytic frontier as the rest of H-LAYER’s deep residual. At second-cumulant order and the operating intensity, the condensate-direction response is certified ($R_{\text{lead}} \leq 0.174, \times 5.7$; prior note), which is the strongest statement available at this order.

6. Status

T3 FORMULATION. The full off-diagonal Hessian is constructed; the Gershgorin combinatorial part is additive-energy controlled (R-025/R-026); the analytic obstruction (exchange-block sign, not fixed by $u_{\text{eff}} > 0$) is isolated and shown to MERGE with RES-5. **OPEN:** the exchange-Hessian eigenvalue bound = RES-5. No tier flip: B2 T6, B1 T6 on {H-LAYER}. **Consequence for the H-LAYER ledger:** RES-1’s off-diagonal axis is now resolved into (a) second-cumulant / operating-intensity: condensate-direction CERTIFIED, and (b) the full worst-direction norm: MERGED with RES-5. The H-LAYER residual thus consolidates to RES-5 (+ off-anchor robustness), with RES-1, RES-3 (lattice), and RES-4 all controlled at their stated scopes.

7. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*You stopped short – compute the exchange eigenvalues.*” VALID, and that is RES-5 by construction: the dressed exchange block is a beyond-second-cumulant object (it resums into the matched-order remainder), so it cannot be closed within the second-cumulant scope of this programme. Honestly named a formulation, not a certificate.
- (β) “*Then the constant certificate was weaker than it sounded.*” PARTLY: the constant certificate bounds the second-order condensate-direction response (the physical Bloch competitor), which is the operative quantity for the Reading-H selection at matched order; it never claimed the full norm. This note makes the boundary explicit.
- (γ) “*Does anything here threaten B1 T6?*” DISMISSED: B1 is T6 CONDITIONAL on {H-LAYER}; H-LAYER is the named hypothesis, and this note only sharpens its residual map (RES-1 off-diagonal $\rightarrow$ RES-5). No tier moves.

Quantitative sanity check (mandatory): *sign* – bare $u = -0.86 < 0$, dressed $u_{\text{eff}} = +2.685 > 0$, so the exchange sign is genuinely unfixed by u_{eff} ; *combinatorics* – BCC Gershgorin max row-sum $48 \leq NE_+ = 6480$ (additive-energy controlled); *consistency* – the second-order $R_{\text{lead}} \leq 0.174$ remains the operative bound at matched order; *scope* – exchange eigenvalues are beyond second cumulant, the RES-5 order.

8. Result footer

Result ID:	B2-PROPA-HLAYER / H-diag discharge IV (full operator-norm formulation; RES-1 off-diagonal merges with RES-5)
Precise statement:	Hess $F G^* = E + B_{\text{od}}$; the worst-direction certificate $ E^{-1/2} B_{\text{od}} E^{-1/2} < 1$ needs the EXCHANGE (Fock) block eigenvalues, whose sign is NOT fixed by $u_{\text{eff}} > 0$ (bare $u = -0.86 < 0$; $u_{\text{eff}} > 0$ only via sextic dressing). The Gershgorin row-sums are additive-energy controlled (R-025/R-026; BCC max row-sum $48 \leq N E_+$), but the analytic exchange-sign input is a beyond-second-cumulant object $\Rightarrow$ the full norm MERGES with RES-5. At second cumulant + operating intensity the condensate-direction response is certified ($R_{\text{lead}} \leq 0.174$, prior).
Scope:	Second-cumulant; the full norm is explicitly out of scope (RES-5). Formulation, not certificate.
Dependencies:	hdiag-fullcovariance-formulation (F0 convex, u_{eff}); hdiag-offdiag-additive-energy (sum $p^2 = E_+$); hdiag-offdiag-constant-certificate ($R_{\text{lead}} \leq 0.174$); Math428 (W,B); R-025/R-026.
Evidence grade:	ANALYTIC (formulation) + EXECUTED (hdiag_gershgorin_rowsum.py 3/3).
Reproduction command:	python codes/vacuum/hdiag_gershgorin_rowsum.py
Expected output:	max row-sum $48 \leq N E_+$; $u_{\text{bare}} = -0.86 < 0 < u_{\text{eff}} = 2.685$; 3/3.
Falsification gate:	A proof that $B_{\text{od}} \geq 0$ follows from $u_{\text{eff}} > 0$ (would close the norm without RES-5); or an exchange eigenvalue computation giving $. \geq 1$ at operating intensity (would refute the condensate-direction certificate).
Tier before / after:	No tier flip. B2 T6, B1 T6 on {H-LAYER} unchanged. RES-1 off-diagonal: full worst-direction norm = RES-5-merged; second-cumulant condensate-direction certified.
No-overclaim statement:	The full operator norm is NOT closed; it merges with RES-5. This note is a formulation isolating the analytic obstruction (exchange-block sign), not a discharge. B1 stays T6 on {H-LAYER}; the H-LAYER residual consolidates to RES-5 + off-anchor robustness.
Next required action:	RES-5 (matched-order to exact / GAP-2): the dressed exchange-Hessian eigenvalue bound is the same frontier; attack RES-5 directly (third-cumulant / exact-remainder envelope) rather than as an isolated off-diagonal sub-task.